DRNL | High-level workflow

Double-Radius Node Labeling converts local topology around a candidate link into structural node features

1 Candidate link

Select target nodes u and v inside an enclosing subgraph. The target edge is not used for classification.

2 Two distance views

For every node i:
- distance to u with v removed
- distance to v with u removed

3 Distance pair

Represent the node by $(d_u(i), d_v(i))$. This describes its position relative to both targets.

4 DRNL mapping

Let $d = d_u + d_v$, $q = \lfloor d/2 \rfloor$ and $r = d \bmod 2$.

$$z(i) = 1 + \min(d_u, d_v) + q(q + r - 1)$$

5 Structural feature

Assign one integer label z(i), then one-hot encode it. DGCNN receives these labels as node features.

KEY PROPERTIES

Targets	z(u) = z(v) = 1	**Unreachable**	z(i) = 0	**Input**	local topology only	**Output**	permutation-invariant structural roles

Exact mapping used by SEAL_original · util_functions.node_label()